

ECON 101: Macroeconomics

Solutions – Quiz 5

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Part A — Multiple Choice Solutions

1. A decrease in Canadian interest rates will:

Correct answer: (B) Shift AD right.

Reasoning:

- Lower interest rates reduce the cost of borrowing.
- This stimulates investment spending by firms and durable consumption by households.
- In the AD–AS model, higher planned spending at each price level shifts the *aggregate demand curve* to the right.
- It is not a movement along AD (price level has not changed) and does not directly shift SRAS.

2. Which of the following would shift the SRAS curve to the left?

Correct answer: (C) An increase in corporate taxes.

Reasoning:

- Short-run aggregate supply depends on input costs (wages, raw materials), productivity, and factors that affect profitability.
- An *increase in corporate taxes* raises firms' costs and reduces after-tax profits.
- Firms are less willing to supply output at each price level, so SRAS shifts *left*.
- A decrease in wages and a fall in oil prices reduce costs and would shift SRAS right.
- A technological improvement increases productivity and also shifts SRAS right.

3. The long-run aggregate supply (LRAS) curve:

Correct answer: (C) Is vertical at potential GDP.

Reasoning:

- In the long run, output is determined by *real factors*: labour, capital, technology, and institutions.
- The price level does not affect the full-employment level of output.
- Therefore the LRAS curve is vertical at potential GDP (Y^*).
- It does not slope upward; that describes SRAS.

4. The money multiplier equals:

Correct answer: (B) $\frac{1}{r_d}$.

Reasoning:

- In the simple deposit expansion model, the *simple deposit multiplier* is:

$$m = \frac{1}{r_d},$$

where r_d is the desired reserve ratio.

- The other options either invert or modify this expression incorrectly.

5. If the Bank of Canada sells government securities in the open market, the most likely outcome is:

Correct answer: (B) Higher interest rates and lower AD.

Reasoning:

- When the Bank of Canada *sells* bonds, commercial banks and the public pay for these bonds.
- This *reduces* bank reserves and the money supply.
- With a smaller supply of money, interest rates tend to rise.
- Higher interest rates discourage investment and some consumption, decreasing aggregate demand.

Part B — Numerical / Analytical Problems

Problem 1 (7 marks) — AD–AS & Short-Run Equilibrium

Given:

$$Y^* = 2,500 \text{ (billion dollars)}, \quad P = 120, \quad \Delta I = -50 \text{ (billion)}, \quad MPC = 0.75.$$

Assume the price level is fixed for the multiplier calculation.

(a) Compute the expenditure multiplier. (1 mark)

The simple spending (Keynesian) multiplier is:

$$k = \frac{1}{1 - MPC}.$$

Substitute $MPC = 0.75$:

$$k = \frac{1}{1 - 0.75} = \frac{1}{0.25} = \boxed{4}.$$

(b) Calculate the total change in real GDP. (2 marks)

Initial change in autonomous spending:

$$\Delta A = \Delta I = -50 \text{ (billion)}.$$

Total change in equilibrium real GDP:

$$\Delta Y = k \cdot \Delta A = 4 \times (-50) = \boxed{-200 \text{ (billion)}}.$$

Interpretation: A \$50 billion fall in investment results in a \$200 billion fall in equilibrium GDP.

(c) What is the new level of real GDP? (1 mark)

Initial real GDP was at potential:

$$Y_0 = Y^* = 2,500.$$

After the shock:

$$Y_1 = Y_0 + \Delta Y = 2,500 + (-200) = \boxed{2,300 \text{ (billion)}}.$$

(d) AD–AS short-run effects on GDP, unemployment, and price level. (3 marks)

Step-by-step explanation:

- The reduction in investment shifts the *aggregate demand* curve leftward (from AD_0 to AD_1).
- In the short run, SRAS is upward-sloping and LRAS is vertical at $Y^* = 2,500$.
- The new short-run equilibrium is at the intersection of AD_1 and SRAS.
- As a result:

- **Real GDP** falls from 2,500 to 2,300: the economy is in a *recessionary gap*.
- **Unemployment** rises because firms reduce output and hire fewer workers.
- **Price level:** In the full AD–AS model, the fall in demand also puts downward pressure on the price level.
- In this question, we used a fixed price level for the multiplier calculation, but conceptually, in the AD–AS diagram, both real GDP and the price level fall in the short run.

Problem 2 (8 marks) — Simple Deposit Multiplier

Given:

$$\text{Initial new deposit} = \$5,000, \quad r_d = 0.10.$$

(a) Compute the simple deposit multiplier. (1 mark)

Formula:

$$m = \frac{1}{r_d}.$$

Substitute $r_d = 0.10$:

$$m = \frac{1}{0.10} = \boxed{10}.$$

(b) Maximum possible increase in total deposits. (2 marks)

The maximum total change in deposits is:

$$\Delta D_{\max} = m \times \text{initial new deposit}.$$

$$\Delta D_{\max} = 10 \times 5,000 = \boxed{\$50,000}.$$

Interpretation: If banks lend out all excess reserves and all loans are redeposited in the banking system, total deposits can rise by \$50,000.

(c) Required reserves and loans from the initial deposit. (3 marks)

Step 1: Compute desired reserves from initial deposit.

$$\text{Desired reserves} = r_d \times \text{initial deposit} = 0.10 \times 5,000 = \boxed{\$500}.$$

Step 2: Compute excess reserves (amount that can be loaned out).

$$\text{Excess reserves} = \text{initial deposit} - \text{desired reserves} = 5,000 - 500 = \boxed{\$4,500}.$$

Interpretation: From the initial \$5,000:

- The bank must keep \$500 as reserves.
- It can make new loans of up to \$4,500, which then get redeposited and re-lent through the system, generating the full deposit expansion (up to \$50,000 in total deposits in the simple model).

(d) Why the actual increase may be smaller than the maximum. (2 marks)

Any one of the following (with explanation) is acceptable:

- **Cash leakage:** Some of the loan proceeds may be held by the public as cash rather than redeposited, breaking the multiplier chain.
- **Excess reserves:** Banks may choose to hold excess reserves above the required/desired level, so they do not lend out the full amount of excess reserves.
- **Credit risk / weak loan demand:** In a recession or uncertain environment, banks may be reluctant to lend or borrowers may be reluctant to take on new loans.

For full marks, the student should state clearly that these behaviours reduce the effective multiplier, so the actual increase in deposits and the money supply is *less* than the simple deposit multiplier predicts.

End of Solutions